

ICRA 2026 Handover Competition Report

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Abstract—In this report, we present our entry for the Human-to-Robot Handover Competition at ICRA 2026. Our method uses YOLOv8-based object detection and stereo triangulation to estimate the cup’s center point. Using this information, we compute the grasp pose by positioning the gripper at the cup’s centroid, enabling a natural approach motion. The cup’s center is tracked using an Unscented Kalman Filter (UKF). The estimated motion uncertainty is represented by the covariance matrix P , which is used to construct a virtual hyper-ellipsoid for robot risk awareness. In addition, a force-torque (FT) sensor is attached to the robot’s finger to measure the contact force and maintain the cup’s shape during grasping. The robot is controlled in the workspace using a virtual mass-spring-damper model to achieve a smooth approach to the target pose. Finally, the robot is controlled using workspace control through the UR5e Real-Time Data Exchange (RTDE) interface.

I. INTRODUCTION

Human-robot collaboration is one of the most important factors in increasing productivity. Therefore, the robots must be able to operate autonomously. The Robotic Grasping and Manipulation Competition committees have been holding the human-to-robot handover competition to push the research progress in such a field. The robotic handover competition rules are described in [1].

In this report, we present the method used during the competition. Our approach builds upon our 2024 competition system with two major enhancements: a risk-awareness framework and a virtual mass-spring-damper model for end-effector motion control.

II. METHODOLOGY

Our method consists of four main components: *Robot Perception*, *Risk Awareness*, *End-Effector Motion*, and *State Machine*. The hardware setup comprises a UR5e robotic manipulator, a Robotiq 2F-85 gripper, a finger-mounted force-torque (FT) sensor, and a ZED 2i stereo camera.

A. Robot Perception

The camera RGB image (Zed2i stereo calibrated, synchronized, 640x480, 30fps) is passed through the YOLOv8 [2] semantic segmentation network to produce a binary mask of the detected object. Next, the center points on the image plane are determined using the estimated centroid of the mask image. Each point is tracked using the Unscented Kalman Filter. Finally, the 3D position, with respect to the Camera 1 coordinate, is recovered using the DLT stereo triangulation method [3].

The final grasp pose (H_{grasp}) is calculated as the current grasp poses (H_c) with an additional rotation α_u around the gripper’s y -axis denoted as $H_y(\alpha_u)$.

The additional rotation is calculated as:

$$\begin{aligned} A &= [p_{c,x}^g \ p_{c,z}^g]^T \\ B &= [0 \ 1]^T \\ \alpha &= \cos^{-1}\left(\frac{A \cdot B}{\|A\| \|B\|}\right) \\ \alpha_u &= \text{sgn}(p_{c,x}^g) * \alpha \end{aligned} \quad (1)$$

The final grasp pose is given as:

$$\begin{aligned} H_c &= \begin{bmatrix} 0 & 0 & 1 & p_{c,x}^{base} \\ 1 & 0 & 0 & p_{c,y}^{base} \\ 0 & 1 & 0 & p_{c,z}^{base} \\ 0 & 0 & 0 & 1 \end{bmatrix} \\ H_{grasp} &= H_c H_y(\alpha_u) \end{aligned} \quad (2)$$

B. Risk Awareness

We formulate a simple robot risk-awareness model using the covariance matrix (P) and the object velocity (v) estimated by the UKF. The intuition is that if the object is still moving or the UKF uncertainty remains large, the robot should delay its approach. Therefore, (P) and (v) are used to construct a virtual barrier around the object. As the object becomes more stable, the barrier gradually shrinks, allowing the robot to approach more safely.

The barrier model is defined by (B), as shown in the following equation. The *Ellipsoid* equation computes the semi-axis lengths of the ellipsoid from the covariance matrix and scales them according to the current object velocity ($\dot{x}$) and the weighting factor (w).

$$B \leftarrow w \dot{x} \text{Ellipsoid}(P, n_{std}) \quad (3)$$

C. End-Effector Motion

The motion of the robot’s end-effector is modeled after a virtual mass-spring-damper tuned to critically damped for tracking the object center and smooth approach. The controller is as follows:

$$m\ddot{x} + c\dot{x} + k(x - x_d) = 0. \quad (4)$$

where x is the current end-effector’s position, x_d is the target object position, $\dot{x}$ is the current end-effector’s velocity. m , c , k are the virtual mass, damper, spring parameters respectively. Using the explicit Euler integration, we obtain:

$$\begin{aligned} \ddot{x} &= \frac{-k(x - x_d) - c\dot{x}}{m}, \\ \dot{x}_{k+1} &= \dot{x}_k + \ddot{x}_k \Delta t, \\ x_{k+1} &= x_k + \dot{x}_{k+1} \Delta t. \end{aligned} \quad (5)$$

The end-effector's position x_{k+1} is then send to robot control system for tracking.

D. State Machine

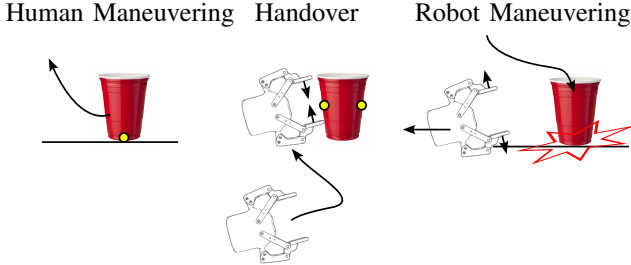


Fig. 1: State Machine

Figure 1 shows the state machine cycle of the task. The following steps are involved in the process:

1. Human maneuvering: At this stage, the cup is placed on the table, and the robot is in a ready position with its gripper open. The human is instructed to pick up the cup for handover. The position data of the cup's bottom (p_b) is used to determine whether it is on the table or in the air.
2. Handover: Once the cup is lifted into the air and becomes relatively stationary, we calculate the grasp pose using the method described in *Robot Perception*. Given the grasp pose, the robot tracks the target in the workspace using the *End-Effector Motion* controller. The *Risk Awareness* model enables the robot to approach the object cautiously, delaying the approach until the object is sufficiently stable. Finally, the gripper closes until contact with the cup is detected by the finger-mounted force-torque (FT) sensor. The robot then moves toward the cup using the UR5e Real-Time Data Exchange (RTDE) library [4]. Then, the human releases the cup.
3. Robot maneuvering: The robot moves the cup toward the desired placing position. We use the "moveUntilContact" method to move the cup until it makes contact with the table. Finally, the gripper is opened, safely retracted from the cup, and the robot is moved to a ready position.

E. Result

Figure 2 shows different poses (left, center, and center) of handover between human and robot in the workspace relative to the robot's base. The method displays robustness to different positions since the IK results are suitable for execution.

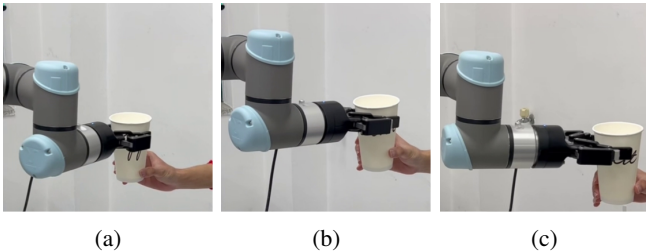


Fig. 2: Handover position: (a) left (b) center (c) right.

III. CONCLUSION

In this report, we presented our approach to the Human-to-Robot Handover Competition for robotic grasping and manipulation. Our method extends our 2024 system with several improvements, including a risk-awareness model based on a velocity-scaled covariance matrix and a virtual mass-spring-damper model for end-effector motion control.

However, several limitations remain. The object detection network struggles to recognize transparent cups or cups containing other objects. During the development of our competition system, we found that perception is the primary bottleneck. As the variety of competition objects increases, the perception module becomes less reliable, resulting in missed detections and an inability to complete the full task.

Future work will focus on improving the robustness of the perception module so that it can reliably detect and recognize a wider range of objects under varying environmental conditions.

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